DS 601 - Mathematical Foundation for Data Science.

Data (what is)

- → Information
- → features of an entity
- → Raw facts
- → Collection of values
- → Computation
- → observation
- → Representation

entity →
- → Prediction ✓ → Evidence
- → Decision
- → Insights
- → Interpretation
- → Patterns
- → Store / Process
- → Conclusions / Decision.

Knowledge

I give you IAB data.

Data → Information → Knowledge

What is Science

- Relationship (Cause & Effect)
- Conducting Experiments
- Establishing logic
- Explain

Correlation ≠ Causation

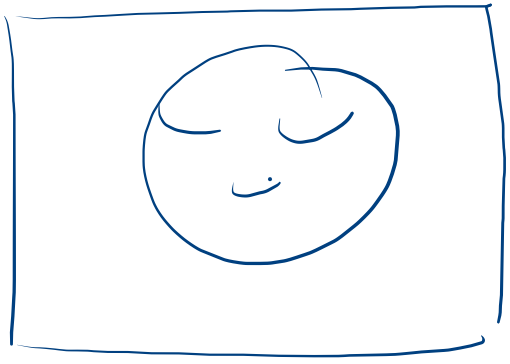
Data Scientist

- Can build Model
- Articulate
- Communicator
- Decision Making
- Mathematical Foundation

Data Analyst

- Data cleaning
- Presentation
- Visualization

Student ID	CPI
001	8.2
002	8.9
003	7.5



Data → Audio, Signals, Text, Image, numbers, Videos Charts.

Data → Collected, Stored, Measured, Visualized, Analyzed

Statistics

Collecting, Samples vs Population

- Bias
- Size of Sample
- Imbalance

Statistics → The science of learning about Population from limited observables.

* What is ML

- Building belief System of a Machine
  - * Show Example
  - * Features
  - * Correct Mistakes
  - * Practice

LA | Calculus | Probability/Stats.

2^{10} — 1 kB

2^{20} — 1 MB

2^{30} — 1 GB

2^{40} — 1 TB

Why LA

Lecture - 2

How does computer Stores the data.

Data → Vetor → Matrices.

Why Calculus →

improving the learning through adjustments/optimization.

Reduce error.

- LA → How to represent / manipulate data
- Statistics → Learn from limited observations / data
- Probability → Reason under uncertainty
- Calculus → How do we optimize.

Data

Student ID	2026123
Student Age	22
Blood Group	O+
Height	180 cm
Image	
Movie Rating	1 ★

→ Qualitative

Data types

1) Categorical
2) Ordinal → 1st, 2nd, 3rd
3) Numerical → IDs,
4) Nominal → Colors
5) Discrete → Age in years
6) Continuous → Height
7) Quantitative → Height
8) Qualitative

ML is the study of algorithms that

improves automatically through Experience (data)

Events
Sample Space
→ Probability → Quantify uncertainty
Study

Sample Space → a Set of all possible outcomes of an experiment (Some outcomes)

* outcome of a trial is uncertain

Experiment → Always uncertain →
* Can be infinitely repeated
* Well defined Set of possible outcomes

Random Exp → outcome cannot be predicted with certainty.

Keywords
- outcome
- Observation
- Experiment
- hypothesis
- Result

Event – Subset of the Sample Space (S)

~~outcome~~ We say event E occured if actual outcome is in E.

Lets do an Experiment

Suppose you take two random exams and each exam has 5 questions with 2 options (T/F). Correct – 1, Incorrect → 0

* you have to compulsorily take both the exams.

a) What is the sample space of possible scores

Combinations

Set ⟷ Sample Space
Subset ⟷ Event (A)
Elements ⟷ outcomes

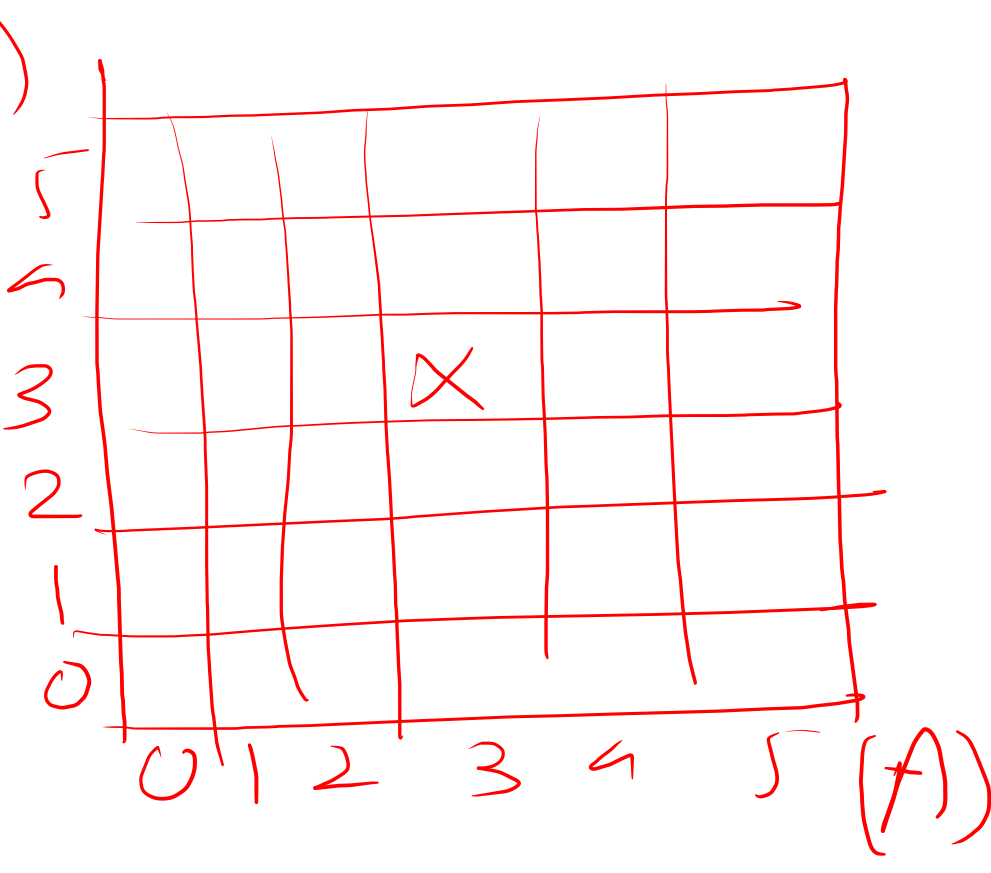
Complement of A

$$A^c = S - A$$

$$x \in A$$

25	54
36	64
72	32

Swan A	Swan B
0	0
1	1
2	2
3	3
4	4
5	5



0, 0	1	0	
1		1	
2		2	
3		3	
4		4	
5		5	

$$P(E) = \frac{\text{No of favourable outcomes}}{\text{No of possible outcomes}}$$ → Event → Sample Space

* Probability of scoring 1

(0,1) (1,0)

$$P(\text{score}=1) = \frac{2}{36}$$

Assumption

1) Equally likely outcomes
2) Finite sample space (36)

} Biased VS Unbiased } Model

Suppose you are taking the exam

1) day

2) Night.

A possible set of Scores from two exams

↓

odd number

$$P(E) = \frac{18}{36}$$

Lecture - 3

Keywords

Set 1) Sample Space $\Omega : \{H, T\}$ S

Subset 2) Event → of our interest

3) outcome

4) Experiment

Set : {1, 2, 3, 4, 5, 6}

No of Subsets $\phi = 2^6$

Cardinality of the Set

$|\Omega| = 6$

$2^6 = 64$

$$P(E) = P = \frac{\text{No of favourable outcomes}}{\text{Sample Space}}$$

Set theory	Probability
Set	Sample space
Subset	Events
elements	outcomes

Principles of counting

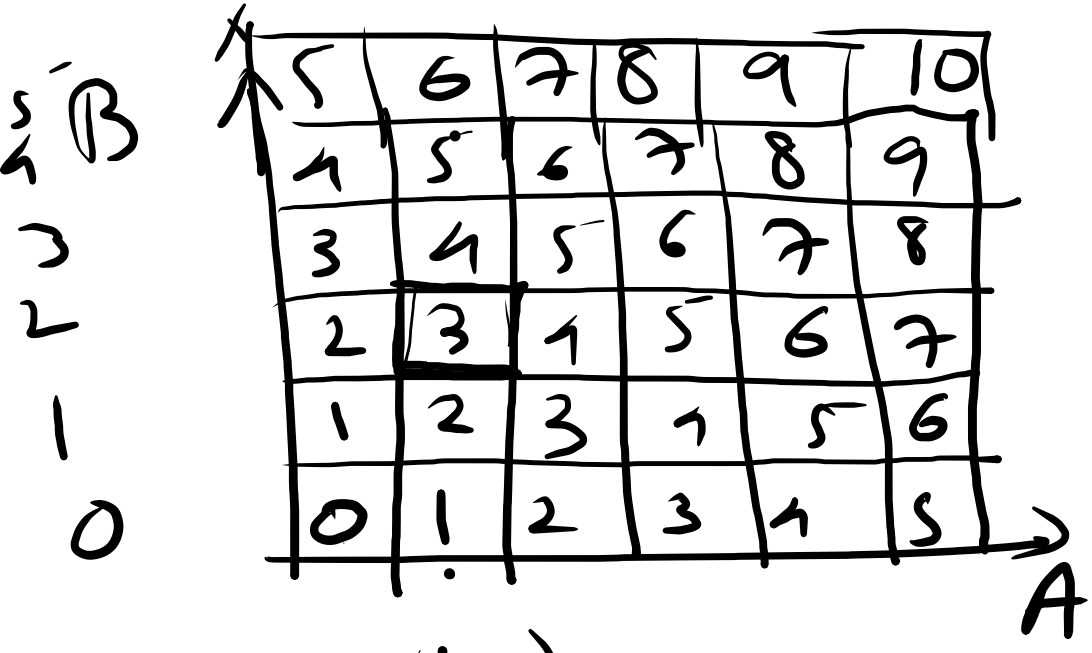
(And) Multiplication Rule & Addition Rule (OR)

Consider an experiment with n_1 possible outcomes, now suppose there is a 2nd exp.

And for each outcome of 1st exp there are n_2 outcomes for the 2nd exp.

$n_1 = 6$ & $n_2 = 6$

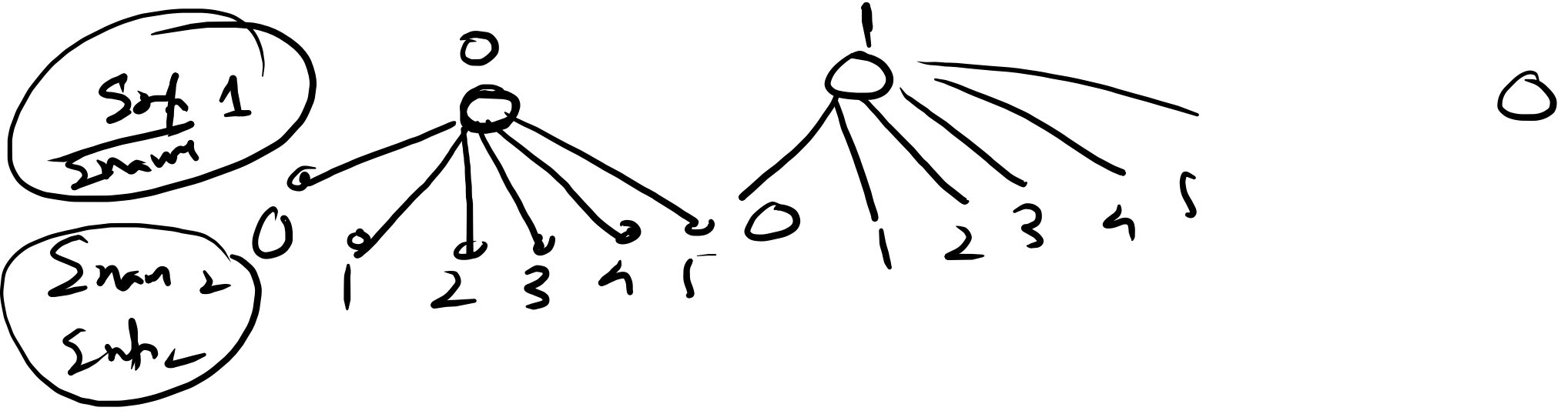
Overall possible outcomes $= 36$



$P(\text{Even Score}) = \frac{18}{36}$

Recall the even numbers

$P(\text{odd Score}) = \frac{18}{36}$



Second Scenario → 2 tasks are performed in exclusion. (OR)

Let Say there are 30 Girls & 30 Boys in this class. Possibility of selecting only 1.

30+30=60

* Permutation → Order $n!$

* Combination → Selection ${}^{n}C_{k}$

* Mutual Exclusion → Cannot happen at the same time.

Sampling Table

Choosing k objects out of n

	Order matters	Order does not matter.
Replacement	1	2
Without Replacement	3	4

n^k	
	n_{C_k}

A

Birthday Problem

Model

60 Students & we want to find the probability of 2 student have same birthday

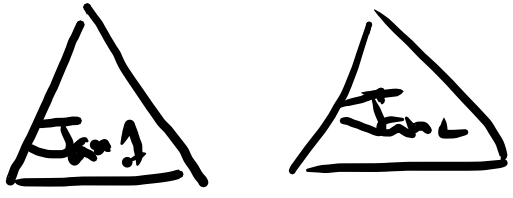
Assumption

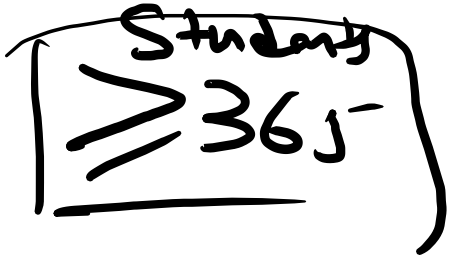
365 days → * All outcomes equally probable likely

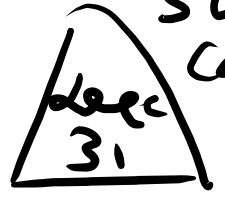
* No consideration twins.

Assume

* Class has 365 Students







Guess
10%.
20%.
0.1
Counter Intuitive

↳ Not an interesting Problem

Let's count the complement → no of possible cases when no 2 students have same birthday.

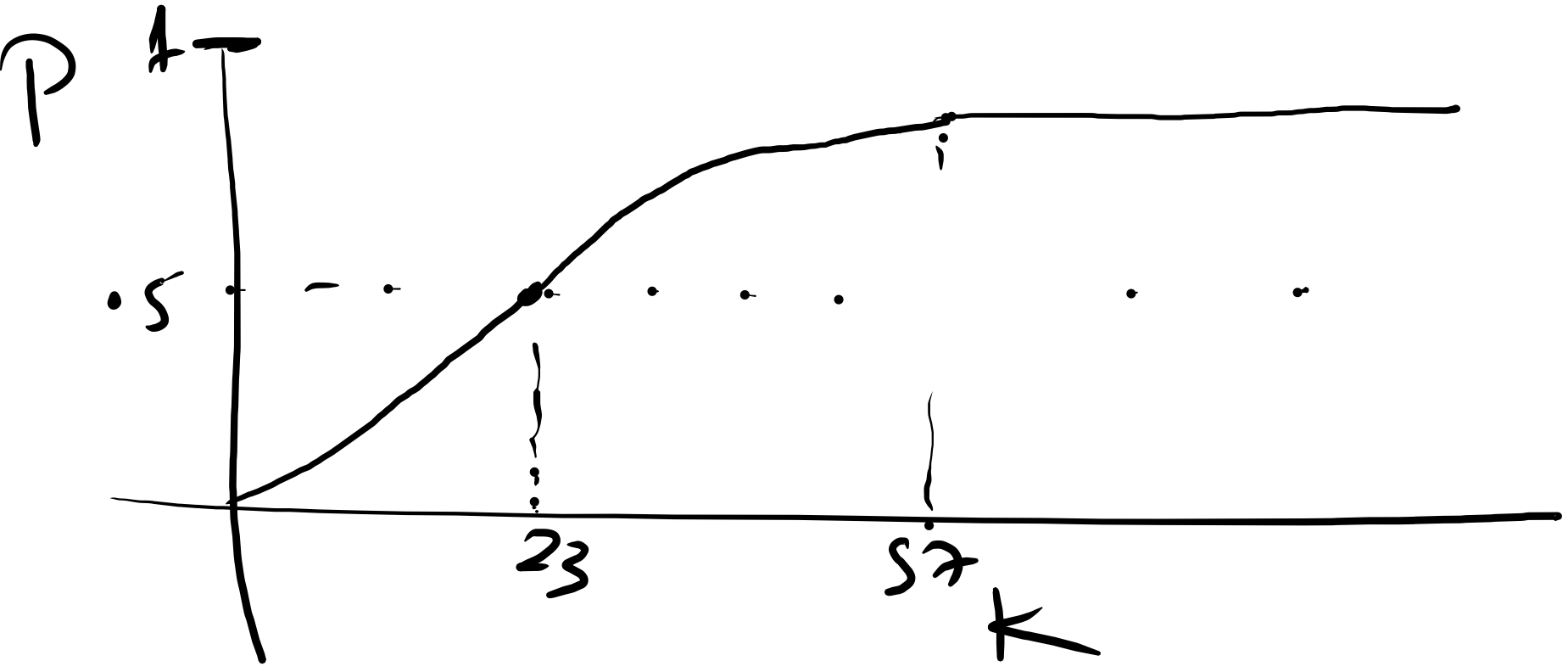
No of possibilities $= 365 \times 364 \times \cdots \times (365-k+1)$

P(no match → compliment)

$= \frac{365 \times 364 \cdots (365-k+1)}{365^k}$

Probability of atleast 1 birthday match

$$P = 1 - \frac{365 \times 364 \cdots (365-k+1)}{365^k}$$



Around class size = 23 P = 50%

" = 57 P = 99%

Q1 60 Student and I want to select 3 Student to answer 3 Questions (one after the other) →

Answer 1 → 60^3 ✓

" 2 → $60 \times 59 \times 58$ ✗

60	60	60

60^3

Q2 choose 3 from 60

$^{n}C_{k}$

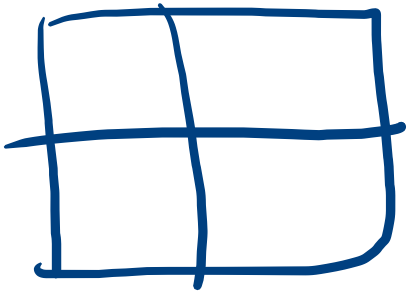
Lecture - 4

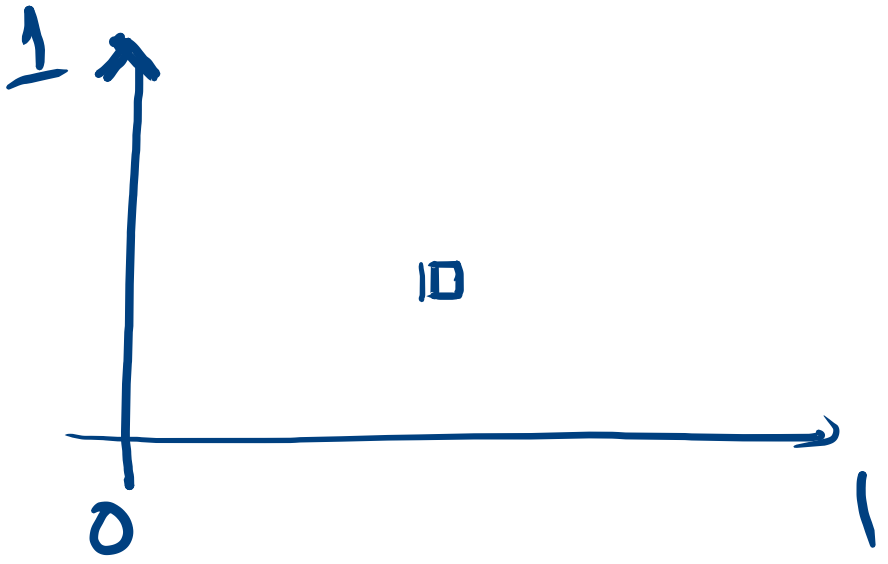
Keywords

1) Principles of counting

2) Birthday Problem

2×2





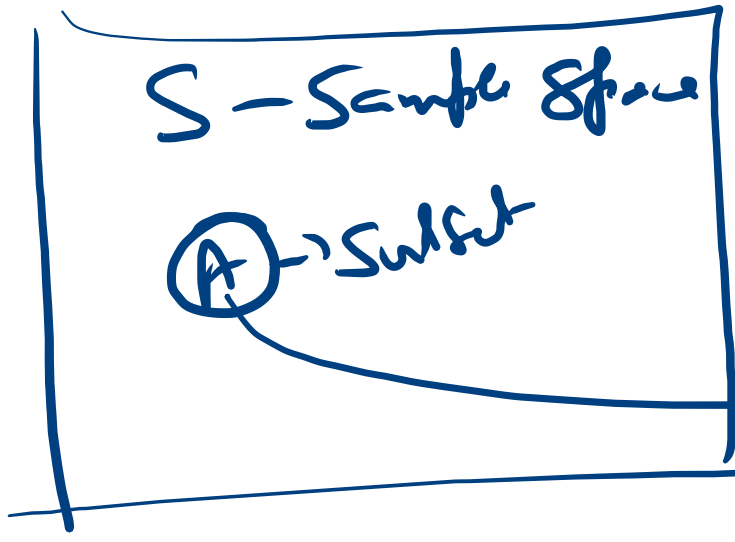
Continuous example

Sample Space
→ Infinite

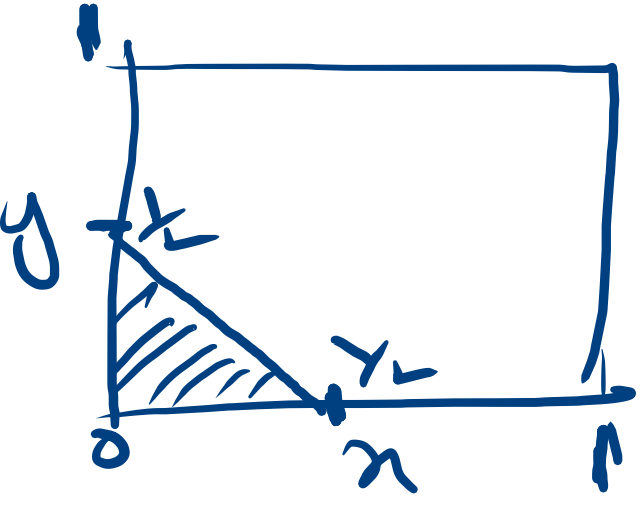
$$P(\bar{t}) = \frac{\text{outcome}}{\text{Sample Space} \to \infty} \to 0$$

✗ Individual probability → 0 → does not mean much

✓ Interested in Subspace.



If the outcome is within this subset then we are interested in that event.



$P(x+y=\frac{1}{2})$

Probability of $(x,y)=(.5,.5)$

Probability of $P(x+y \leq \frac{1}{2})$

$$= \frac{\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}}{1} = \frac{1}{8}$$

Axioms of Probability

Ground rules for probabilistic Modeling.

1) Probabilities should be non-negative ≥ 0

2) Probability of entire Sample Space $= 1 \therefore P(S) = 1$

3) If $A \cap B = \phi$, then $P(A \cup B) = P(A) + P(B)$

$P(A) > 1$

X wrong

$P(S) = 1 = P(A + A^c)$

$P(A + A^c) = 1$

$P(A) + P(A^c) = 1$

$P(A \cap A^c) = 0$

$\therefore P(A) \leq 1$

Properties of Probability from Axioms

(1) $P(A^c) = 1 - P(A)$

(2) A is contained in B

$A \subseteq B$ $\qquad$ $P(A) \leq P(B)$

(3) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

General Inclusion-Exclusion formula

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$

$$P(A_1 \cup A_2 \cup \cdots A_n) = \sum_{i=1}^{n} P(A_i) \ominus \sum_{i<j} P(A_i \cap A_j) \oplus \sum_{i<j<k} P(A_i \cap A_j \cap A_k) \cdots \boxed{?} \ominus \sum_{i_1 < \cdots < i_n} P(A_{i_1} \cap \cdots \cap A_n)$$

Assignment.

The Matching Problem

Gambling ← → Real practical Examples

Deck of N cards — 1 No. Per Card

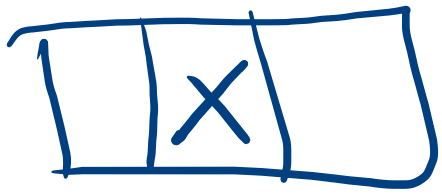
labels are 1 to N

Well Shuffled deck of N cards — You flip over the cards one by one saying the numbers 1 through N

You win the game if the no. you say is the same as the no. on the card.

Probability of Winning?

Total cards 1, 2, 3



Probability of Winning

$$P(\text{Win}) = \frac{\text{Total Permutation}}{\text{Total Permutation}}$$

{(1,2,3), (1,3,2), (3,1,2), (3,2,1) (2,3,1) (2,1,3)}

1 match – one of the cards match

how many out of all permutations have one match

how rest of the $n-1$ cards are arranged $\rightarrow (n-1)!$

Equation $\rightarrow$ $n_{C_1}(n-1)!$

$$P(Win) = \frac{{}^{n}C_1 (n-1)! + {}^{n}C_2 (n-2)! + \dots {}^{n}C_3 (n-3)!}{n!}$$

Wrong → ASSignment need to think

Proposed solution

Test it with 3 cards 1, 2, 3

- 1st term ${}^{3}C_1 = 3$ & $(n-1)! = (3-1)! = 2$
- 2nd term ${}^{3}C_2 = 3$ & $(n-2)! = 1$
- 3rd term ${}^{3}C_3 = 1$ & $(n-3)! = 1$

$$P(Win) = \frac{3\times2 + 3\times1 + 1\times1}{6} = \frac{10}{6} = 1.67$$

X

Match	Subsets	Count	
0	{(2,3,1), (3,1,2)}	2	X
1	{(1,3,2), (3,2,1), (2,1,3)}	3	✓
2	{ 3	0	
3	{(1,2,3)}	1	✓

Exhaustive

$P(\text{Win}) = \frac{4}{6}$

$P(\text{Loss}) = \frac{2}{6}$

$P(\text{Win}) + P(\text{Loss}) = 1$

$\frac{4}{6} + \frac{2}{6} = 1$

✓ Correct Solution

$$\therefore P\left(\bigcup_{i=1}^{n} A_i\right) = 1 - \frac{1}{2!} + \frac{1}{3!} - - + -$$

$$(-1)^{n+1} \cdot \frac{1}{n!}$$

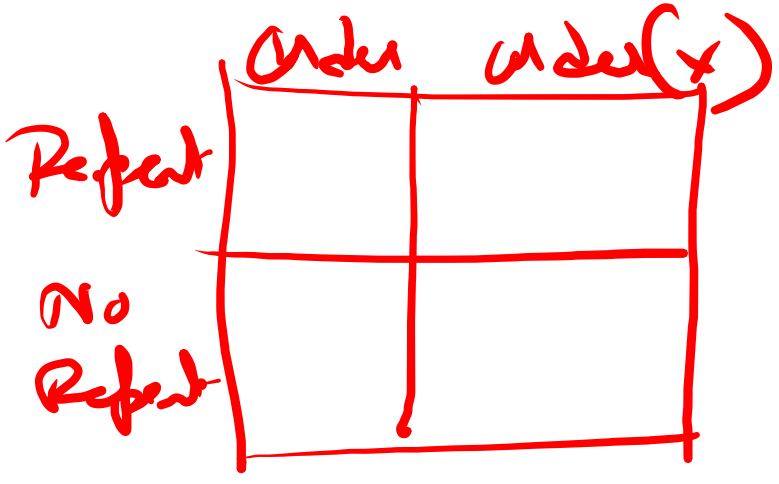
$$= 1 - \frac{1}{e}$$

$$= \frac{n}{n} - \frac{{}^{n}C_2}{n(n-1)}$$

Lecture - 5

Keywords

* Matching Problem → Principles of inclusion/Exclusion



Conditioning (Conditional Probability)

(Heart & Soul of Data Science)

* New information about the experiment
  ↳ change the uncertainties

we revise our model

Independence → Two events A & B are independent.

$$P(A \cap B) = P(A)\,P(B)$$

Disjoint → Mutually Exclusive events

If A, B, C are independent

$$P(A \cap B) = P(A)\,P(B)$$

$$P(A \cap C) = P(A)\,P(C)$$

$$P(B \cap C) = P(B)\,P(C)$$

$$P(A \cap B \cap C) = P(A)\,P(B)\,P(C)$$

Newton - Pepy's Problem (Invalid Gambling)

* Have fair dice, which is most likely to happen

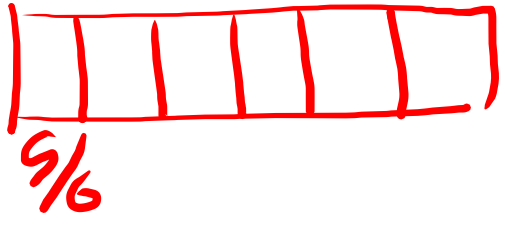
Event(A) → With 6 rolls, <u>at least</u> one 6

Event(B) → " 12 " , " " 2 6 Sin's

Event(C) → " 18 " , " " 3 Sixes.

P(A) = 1 − Probability of no Six

= $1 - \left(\frac{5}{6}\right)^6$



} 6 rolls are independent

$$P(B) = 1 - \left(\frac{5}{6}\right)^{12} - 12\left(\frac{1}{6}\right)^{1}\left(\frac{5}{6}\right)^{11}$$

(12)

↓ Previous

↳ Probability of 1st six

↳ All flaws where can come

$$P(C) = 1 - \sum_{k=0}^{2} \binom{n}{k} \left(\frac{1}{6}\right)^{k}\left(\frac{5}{6}\right)^{n-k}$$

$P(A) \approx .665$ $\quad P(C) \approx .6$

$P(B) \approx .619$

Conditional Probability

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

→ Probability of A when B occurs

$$P(A \cap B) = P(B)\ P(A/B)$$

$$P(B \cap A) = P(A)\ P(B/A)$$

$$P(A \cap B) = P(B \cap A)$$

$$P(B)\ P(A/B) = P(A)\ P(B/A)$$

$$P(A/B) = \frac{P(A)\ P(B/A)}{P(B)}$$

Conditional Thinking

Question A patient comes and get tested for covid and suppose it affects 1% of population.

Suppose test is 95% accurate.

* (what is the P(s he has covid).

Event D → Patient has disease

event T → " tests true

$P(T/D) = .95$ (no use)

$P(T^c/D^c) = .95$ "

$P(A/B)$
$P(B/A)$
$P(A/E)$
~~$P(E/A)$~~

$P(D/T)$ whether patient has the disease

$$P(D/T) = \frac{P(T/D)P(D)}{P(T)}$$

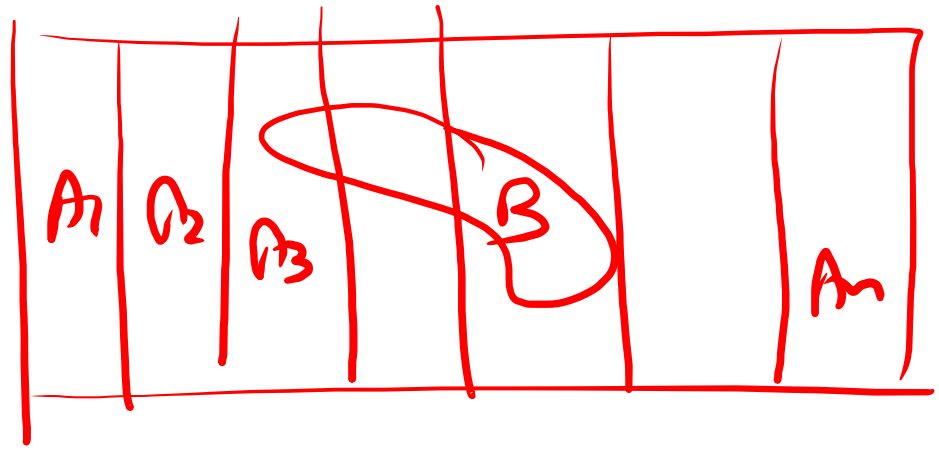
which of these probabilities are known

$P(D) \rightarrow 1\%$

$P(T) \rightarrow$? do we need this number.

General Strategy → Breaking Problem into Simple pieces.

Law of total Probability



Find $P(B) = ?$

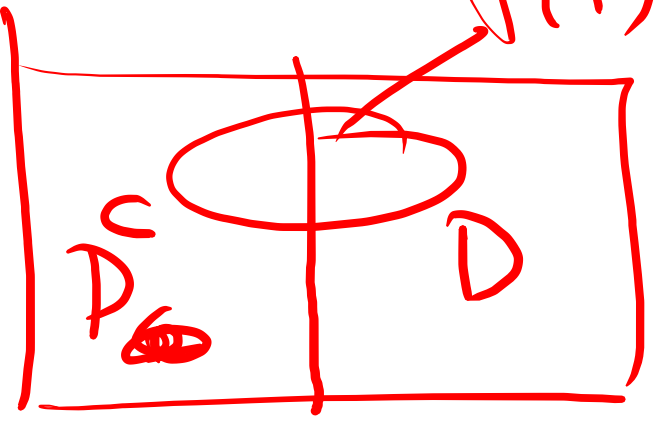
$$P(B) = P(B \cap A_1) + P(B \cap A_2) \cdots P(B \cap A_n)$$

$$= P(B/A_1)P(A_1) +$$

Unconditional Probability — Conditional Probability

$$P(D|T) = \frac{P(T|D)\,P(D)}{P(T)}$$

$$= \frac{P(T|D)\,P(D)}{P(T|D)\,P(D) + P(T|D^c)\,P(D^c)}$$



$$= \frac{.95 \times .01}{.95 \times .01 + .05 \times .99}$$

$$\approx 16\%$$

$P(D) = .01$

$P(T|D) =$

Sample Space → 1000

1% → 10 ⟶ All 10 will test +ve

99% → 990 ⟶ 50 will test +ve

information is interesting

Lecture - 6

Keywords
- Conditional Probability
- Bayes Theorem
- Total Probability

Solved completely information

No update

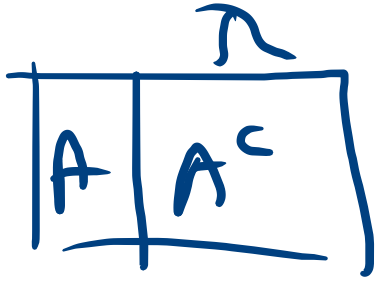
Partial update

Mutually Exclusive

$P(A \cap B) =$

A & B are independent

Interested in event A



Some information about event B

$P(A) = P(A|A)$

$P(A) =$

$P(A|B) =$

Sure (Certain)

$P(A/A) = 1$

(Uncertain)

$P(A)$ → Prior

$P(A/B)$ → Posterior

Back to the covid Problem

1% population is affected

95% accurate (test)

Person tested true

✗ No particular Symptoms

H → Hypothesis — patient has the disease

E → Evidence (Event) — test is true

P(H) → Probability that hypothesis is true

patient has the disease without event → 1 - /.

P(E) → Probability of seeing the evidence

~~Probability of testing true~~ test is accurate → 95 /.

P(H/E) → Given the evidence, probability that hypothesis is true (

P(E/H) → If the hypothesis is true, probability of seeing the evidence. (if patient has disease, probability that tested with test

$$P(H|E) = \frac{P(H)\,P(E|H)}{P(E)}$$

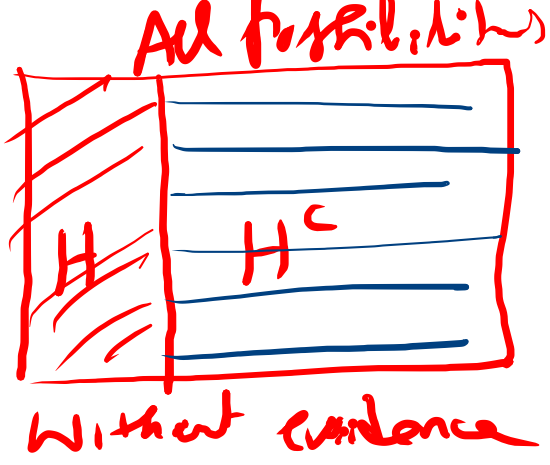
$P(E|H)$ ✗

$$= \frac{}{P(H) * P(E|H) + P(H^c)\,P(E|H^c)}$$

1st term 2nd term

1st term → probability of having disease & testing true

2nd " → " " not having disease & testing true (falsely identified)





$$P(H/E) = \frac{.01 \times .95}{.01 \times .95 + .05 \times .99} \approx 16\%$$

2nd step — Second opinion (test from 2nd lab)

$P(H) \to 16\%$

$$P(H/E) = \frac{.16 \times .95}{.16 \times .95 + .05 \times .84}$$

$P(H/E)$ → 2nd

$\approx 78\%$

95% accurate test → Pretty bad

Common Mistakes

(1) Confusing $P(A|B) = P(B|A)$

$P(\text{Ill}|\text{test}) = P(\text{test}|\text{Ill})$ →

Confusing (Prosecutor's Fallacy)

(2) Confusing Prior with Posterior

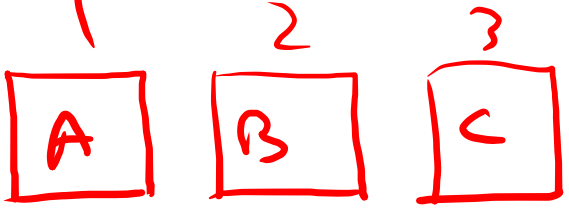
$$P(A) = P(A|A)$$

(3) Confusing independence with conditional independence.

Lecture - 7

13 Aug

Game Show
Monty is the host



3 doors
Pick a Door
Car behind one door
Two doors have goats

Assumption - 1. You don't know where the car is but Monty knows.

2 You want the car, not the goat

3. Monty always opens a door where goat is there, & car is not there - If he has a choice, he picks with equal probability.

You picked door A $p = \frac{1}{3}$

Monty has opened B → There is a goat behind door B

Keywords

Bayes' Theorem

Common Misconception in probability

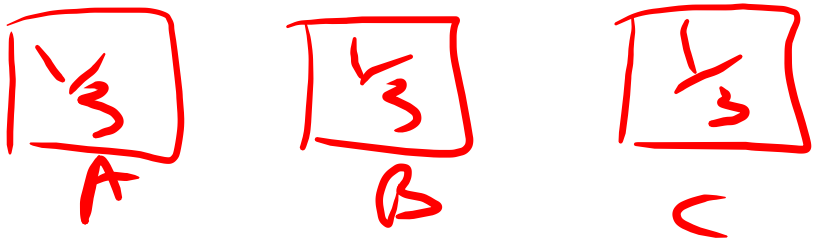
1) Door B has goat
2) Monty opened door B

} 2 evidence

Prior	Evidence: B open	Posterior
$P(\text{car@A}) = \frac{1}{3}$	$P(\text{open B} \mid \text{car A}) = \frac{1}{2}$	$P(\text{car@A} \mid \text{B open}) = \frac{1}{3}$
$P(\text{car@B}) = \frac{1}{3}$	$P(\text{open B} \mid \text{car B}) = 0$	$P(\text{car@B} \mid \text{B open}) = 0$
$P(\text{car@C}) = \frac{1}{3}$	$P(\text{open B} \mid \text{car C}) = 1$	$P(\text{car@C} \mid \text{B open}) = \frac{2}{3}$

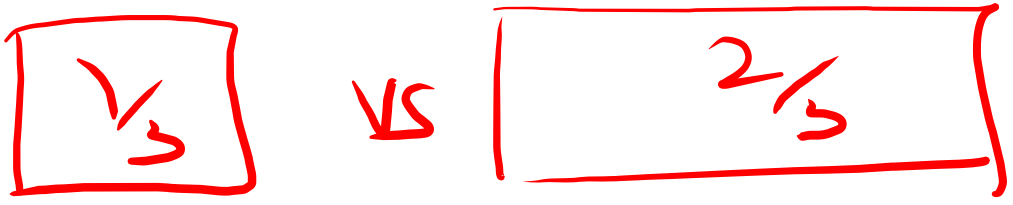
$P(x) = \frac{x}{x+y+z}$

Solution by Visualization

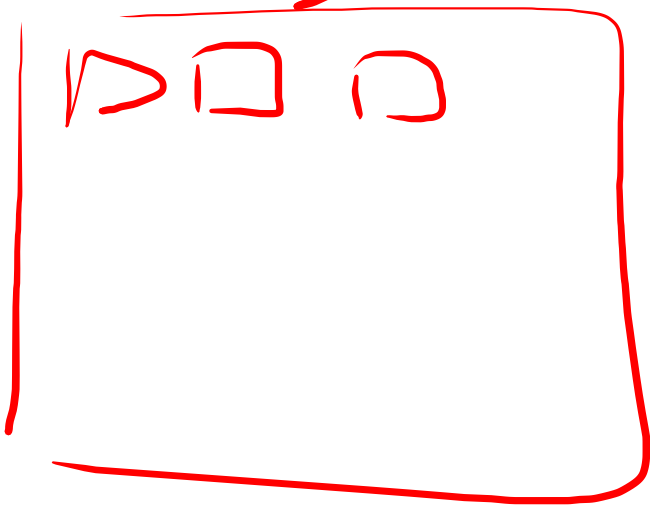


Initial

Suppose you choose A



Sunny



Question

Surgeons

Two doctors - Doctor A vs Doctor B

Doctor A

Heart	Wound
70	10
20	0

S - Success
F - Failure

Costly

Doctor B

Heart	Wound
2	81
8	9

S
F

Charges less

Which doctor will you Recommend?

Doctor A Success rate $= \frac{\text{No. of Success}}{\text{Total}} = \frac{80}{100}$ A or B

$= 80\%$.

B " $= \frac{83}{100} = 83.1$

Categorical Success / Failure

		Surgery	Wound
Surgery	A → Success Rate →	$\frac{70}{90}$	$\frac{10}{10}$
	B → " →	$\frac{2}{10}$	$\frac{81}{90}$

Simpson's Paradox — Phenomena in Probability & Statistics, in which a trend appears in different groups of data but disappears or reverses when these groups are combined (with)

Paradox

Confounding problem (which type of Surgery)

Doctors D_1 - A

D_2 - B

A: Surgery is Successful

B: Treated by doctor D_2

C: Heart Surgery

Treatment Flips

$$P(A|B,C) \leq P(A|B^c,C)$$

$$P(A|B,C^c) \leq P(A|B^c,C^c)$$

} Conditional

Overall

$$P(A|B) > P(A|B^c)$$

} Unconditional